Foundations Notebook — canonical base

The Method of Equal Shares, founded on Peters–Pierczyński–Skowron, applied to Durango

Notation, definitions, and theory re-derived on the rock of

Proportional Participatory Budgeting with Additive Utilities

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Abstract

This notebook re-founds our entire participatory-budgeting (PB) theory on a single source: Peters, Pierczyński & Skowron's *Proportional Participatory Budgeting with Additive Utilities* (arXiv:2008.13276v2; NeurIPS 2021). We adopt their notation and definitions wholesale, reproduce the load-bearing results (the Method of Equal Shares, Extended and Fully Justified Representation, the core, priceability), and then rebuild our Durango contribution on top: the District-Exclusivity Mechanism (DEM), its distortions, and the case that Equal Shares repairs them while delivering geographic equity *by construction*. Remarkably, Peters et al.'s own motivating example—*Circleville*, a city run as separate district elections—is our Durango critique almost verbatim. This is the canonical reference for the project; its section order is the intended skeleton of the paper.

How to read this, and three conventions. (1) Boxes. Green = the load-bearing idea; blue = intuition; teal = the bridge to Durango; *red "Watch out"* = a subtle point that is genuinely easy to get wrong; violet = a label we are coining (so you can always tell ours from the literature's). **(2) Provenance.** The Method of Equal Shares, EJR, FJR, the core, priceability, PAV, PSC, and GCR are *theirs* and are cited. The District-Exclusivity Mechanism, the Unrestricted Allocation Mechanism, district *salience*, the voter *information function*, and π_d*-geographic equity* are *ours* and are flagged in violet boxes. **(3) Notation.** We now follow Peters et al. exactly. Appendix A gives the crosswalk from our old `notebook_v2` symbols; the headline changes are C for projects, b for budget, $\mathrm{cost}(\cdot)$ for cost, plain ρ for the Equal-Shares threshold, and the renamings s_d (salience), κ_i (information), π_d (population share) that free α, σ, β for their roles in Peters et al.

Contents

1 Introduction: from Circleville to Durango

A growing number of cities allocate part of their budget by *participatory budgeting* (PB): residents vote over candidate projects, and a rule turns the ballots into a funded set subject to a budget constraint (Cabannes, 2004; Wampler et al., 2021; Aziz and Shah, 2020). The question this project asks is narrow and practical: *is the rule Durango uses a good one, and if not, what should replace it?*

Peters, Pierczyński & Skowron (2021) open with a fictional city, **Circleville**, split into four districts of unequal size. If projects are local and residents vote only for their own district's projects, then a single city-wide "most-votes-win" count hands the entire budget to the largest district—what they call *the tyranny of the largest minority*. The standard fix is to *run separate elections per district*, dividing the budget in advance (often in proportion to population) and letting residents vote only locally. Peters et al. then catalogue exactly what that fix breaks:

- **Boundary projects are orphaned.** A project on the border of two districts must be assigned to one; residents of the other cannot vote for it, so it is under-funded relative to its true value.
- **City-wide goods are hard to place.** A project benefiting the whole city has no natural district.
- **Non-geographic interest groups are silenced.** Parents, or cyclists, spread across the city cannot form a voting bloc, even when their shared project would be highly valuable.

Connection to Durango

Durango's *presupuesto participativo* **is Circleville's "separate district elections."** A resident selects one district and casts a fixed number of votes (v, e.g. six) *within* that district only. This is precisely the fix Peters et al. diagnose—so their critique is, line for line, our motivation. Our contribution is to (i) make the distortions precise as propositions in their model, with Durango's institutional detail, and (ii) argue that their remedy, the Method of Equal Shares, is the right reform for Durango because it keeps a single city-wide election while protecting marginalized districts *by construction*.

Two features of the Durango setting are not incidental; they shape the whole problem and must stay central.

A name we are introducing (our label, not standard terminology)

The voter information function κ_i (rational ignorance). A voter typically knows only the projects in her immediate circle. Under Durango's rule she must nonetheless spend *all* v votes inside her chosen district—including on projects she has never heard of. We write $\kappa_i \subseteq C$ for the set of projects voter i can actually evaluate. The mechanism forces votes outside κ_i, converting ignorance into noise. (Background: rational ignorance, Downs 1957; costly information in voting, Martinelli 2006.)

A name we are introducing (our label, not standard terminology)

District salience s_d and the centralization risk. Removing the district constraint entirely (our *Unrestricted Allocation Mechanism*, §2) lets high-visibility city-center projects absorb the vote. We capture visibility by a *salience* parameter s_d per district: high-s_d districts attract attention, low-s_d (often marginalized) districts get none. The district constraint is a *blunt equity instrument*—it protects low-salience districts by accident. The reform must preserve that protection deliberately, not discard it.

Key idea

The design target Peters et al. identify, and which we adopt for Durango, is: *a single city-wide election whose rule spends money proportionally*, automatically identifying groups with common interests (geographic or not) and guaranteeing each its fair share. That target is made precise by the axiom *Extended Justified Representation* (EJR) and met by the *Method of Equal Shares.*

The rest of the notebook follows Peters et al.'s architecture: the model (§2), the Method of Equal Shares (§3), the proportionality axioms (§4), then our Durango layer—the DEM's distortions (§5), their repair under Equal Shares (§6), the three-way comparison (§7), and the simulation design (§8).

2 The participatory-budgeting model

We use Peters et al.'s model verbatim, then add the few district-level objects Durango needs. For $t \in \mathbb{N}$ write $[t] = \{1, \dots, t\}$.

Definition 2.1 (Election; general PB model)**.** An *election* is a tuple $E = (N, C, b, \mathrm{cost}, \{u_i\}_{i \in N})$ where

- $N = [n]$ is the set of *voters* and $C = \{c_1, \dots, c_m\}$ the set of *projects* (candidates);
- $b \in \mathbb{Q}_{\geq 0}$ is the available *budget*;
- $\mathrm{cost}\colon C \to \mathbb{Q}_{\geq 0}$ gives each project's cost, extended to sets by $\mathrm{cost}(T) = \sum_{c \in T} \mathrm{cost}(c)$;
- each $u_i\colon C \to \mathbb{Q}_{\geq 0}$ is voter i's *additive* utility, so a funded set T gives i utility $u_i(T) = \sum_{c \in T} u_i(c)$; for $S \subseteq N$ write $u_S(T) = \sum_{i \in S} u_i(T)$. We assume $u_N(c) > 0$ for every c.

A set $W \subseteq C$ is *feasible* if $\mathrm{cost}(W) \leq b$; a feasible W is an *outcome*. An *(aggregation) rule* R maps each election E to an outcome $R(E)$.

Definition 2.2 (Two special cases)**.** *Committee elections / unit costs:* $b \in \mathbb{N}$ and $\mathrm{cost}(c) = 1$ for all c, so W is feasible iff $|W| \leq b$ (a committee of size b). *Approval utilities:* $u_i(c) \in \{0, 1\}$; then $A(i) = \{c : u_i(c) = 1\}$ is i's *approval set*, and $c \in A(i) \cap W$ is a *representative* of i in W.

Watch out — a subtle point that is easy to get wrong

Two utility readings of the *same* approval ballot differ once costs vary. Under *approval utilities* a voter's satisfaction is the *number* of her approved projects that get funded ($u_i(c) = 1$). Under *cost utilities* it is the total *money* spent on them ($u_i(c) = \text{cost}(c)$). Peters et al. note the latter is often closer to the truth (their Paris example: a project 186× more expensive but only 1.15× more popular). Which reading we use is a modeling choice we must make explicit in the simulation (§8).

2.1 Durango-specific objects

To talk about Durango we layer a district structure on top of the general model.

Definition 2.3 (District structure)**.** Let $\mathcal{D}$ be a set of *districts.* A map $\delta\colon C \to \mathcal{D}$ assigns each project to a district, and each voter i resides in a district $d(i) \in \mathcal{D}$. Write $n_d = |\{i : d(i) = d\}|$ for the population of district d, and $\pi_d = n_d/n$ for its *population share.* A pre-divided budget assigns b_d to district d with $\sum_d b_d = b$ (Durango uses $b_d \approx \pi_d b$).

A name we are introducing (our label, not standard terminology)

District salience $s_d \in \mathbb{R}_{>0}$ scales how much attention/turnout district d's projects attract in an *unconstrained* election. **The information function** $\kappa_i \subseteq C$ is the set of projects voter i can evaluate. Both are ours, not Peters et al.'s.

A name we are introducing (our label, not standard terminology)

The District-Exclusivity Mechanism (DEM)**.** Each voter i selects exactly one district $d_i \in \mathcal{D}$ and allocates her v votes among projects in d_i *only.* Within each district, projects are funded in decreasing order of total votes until b_d is exhausted. This is Durango's actual rule and Peters et al.'s "separate district elections."

A name we are introducing (our label, not standard terminology)

The Unrestricted Allocation Mechanism (UAM)**.** The same ballot but with no district restriction: a voter may direct her v votes at *any* projects, and projects are funded in decreasing order of total votes against the single global budget b. This is the naive "most-votes-win" city-wide rule—the one that yields the tyranny of the largest (most salient) minority.

Intuition

DEM and UAM are the two poles Peters et al. warn about: UAM centralizes (high-s_d districts win everything), DEM fragments (the per-district fix that orphans boundary projects, silences city-wide groups, and—via κ_i—forces uninformed votes). The Method of Equal Shares is the third option that avoids both poles.

3 The Method of Equal Shares

A name we are introducing (our label, not standard terminology)

The *Method of Equal Shares* (MES) is **not** ours; it is Peters et al.'s rule (originally "Rule X," Peters & Skowron 2020). We restate it exactly.

Definition 3.1 (Method of Equal Shares)**.** Give each voter an equal share b/n of the budget. Start with $W = \varnothing$ and add projects one at a time. For a chosen c, let $p_i(c) \geq 0$ be i's payment, with $\sum_{i\in N} p_i(c) = \mathrm{cost}(c)$; write $p_i(W) = \sum_{c\in W} p_i(c)$ and the remaining balance $b_i = b/n - p_i(W)$. For $\rho \geq 0$, a project $c \notin W$ is *ρ-affordable* if

$$\sum_{i\in N} \min\bigl(b_i,\ u_i(c)\cdot\rho\bigr) = \mathrm{cost}(c).$$

If no project is ρ-affordable for any ρ, MES terminates and returns W. Otherwise it selects a $c \notin W$ that is ρ-affordable for the *minimum* ρ, with payments $p_i(c) = \min(b_i, u_i(c)\cdot\rho)$.

Key idea

Each voter pays *per unit of her utility*, at a common price ρ, capped at her remaining balance. Projects enter in increasing order of ρ, i.e. in *decreasing order of utility per dollar.* So MES favours cheap projects, projects with many supporters, and projects whose supporters still hold money (those whose preferences are so far least satisfied).

Watch out — a subtle point that is easy to get wrong

We now write the threshold as plain ρ, matching Peters et al. In the old `notebook_v2` it was ϱ (`\mesrho`) to dodge a LaTeX clash; that dodge is retired.

Approval special case. With $u_i(c) \in \{0, 1\}$, c is ρ-affordable iff its cost can be covered by its approvers so that each pays at most ρ: approvers with less than ρ left spend all they have, the rest pay exactly ρ. Cost is split *as equally as possible*—hence "Equal Shares." The first project chosen is the affordable one maximizing *approvers per dollar.*

Computing ρ (Algorithm 1). For $c \notin W$ with supporters $i_1, \ldots, i_t$ (those with $u_i(c) > 0$), sort by $b_i/u_i(c)$ ascending. Supporters who exhaust their balance attain the cap in the first coordinate. Iterating $s = 1, \ldots, t$, the candidate value is

$$\rho \;=\; \frac{\mathrm{cost}(c) - \bigl(b_{i_1} + \cdots + b_{i_{s-1}}\bigr)}{u_{i_s}(c) + \cdots + u_{i_t}(c)},$$

accepted as soon as $\rho \cdot u_{i_s}(c) \leq b_{i_s}$. The rule then picks $c = \arg\min_{c\notin W} \rho(c)$. With running partial sums this is $O(m^2 n \log n)$.

Example 3.1 (Equal Shares, worked (Peters et al.'s instance))**.** Ten voters, budget $b = \$100$ (so each starts with \$10), approval utilities, five projects: P1 \$36 approved by i_1–i_5; P2 \$36 by i_1–i_6; P3 \$25 by i_1–i_5; P4 \$24 by i_4–i_7; P5 \$24 by i_6–i_9. Round 1 thresholds: $\rho_{P1} = 7.2$, $\rho_{P2} = 6$, $\rho_{P3} = 5$, $\rho_{P4} = \rho_{P5} = 6$. MES funds **P3** (lowest ρ); i_1–i_5 each pay \$5. Now P1, P2 are unaffordable (supporters too poor); recomputing, $\rho_{P4} = 7$, $\rho_{P5} = 6$, so MES funds **P5**; i_6–i_9 pay \$6. Nothing else is affordable: $W = \{\mathrm{P3}, \mathrm{P5}\}$, spending only \$49 of \$100. The leftover illustrates *non-exhaustiveness* (§4); completion methods address it.

Connection to Durango

Read the five projects as Durango projects from different districts and the ten voters as a mixed electorate. MES never asked anyone to "spend all six votes in one district." A voter pays only for projects she actually values—exactly what κ_i demands—and a small, cohesive group (e.g. supporters of P5) secures its project out of *its own* share, with no district quota imposed from above.

4 Proportionality: EJR, the core, priceability, FJR

4.1 Why not just maximize welfare? (Theorem 1)

Theorem 4.1 (Costs matter; Peters et al., 2021)**.** *Every rule that depends only on voters' utility functions and on the collection of budget-feasible sets must fail proportionality—even on instances with a clean district structure.*

Intuition

Onetown vs. Twotown. Two cities with identical voters, districts, and approval sets, and even identical feasibility constraints, but different *costs*. Proportional Approval Voting (PAV), which maximizes $\sum_i H(|A(i) \cap W|)$ with H the harmonic number, must return the same outcome in both—yet the proportional outcome differs between them. Hence any welfare-optimizing rule that ignores costs cannot be proportional once costs vary. This is why we cannot simply "run PAV" on Durango, and why a *cost-aware, sequential* rule like Equal Shares is needed.

4.2 Extended Justified Representation (EJR)

EJR formalizes "no cohesive group is under-served." Peters et al. build it in three steps.

Definition 4.1 (EJR, committee/unit-cost; Aziz et al., 2017)**.** A group S is *ℓ-cohesive* if $|S| \geq \frac{\ell}{b} n$ and $\left|\bigcap_{i \in S} A(i)\right| \geq \ell$. A rule satisfies *EJR* if for every ℓ and every ℓ-cohesive S there is $i \in S$ with $|A(i) \cap R(E)| \geq \ell$.

Definition 4.2 (EJR, approval, general costs)**.** S is *T-cohesive* ($T \subseteq C$) if $|S|/n \geq \mathrm{cost}(T)/b$ and $T \subseteq \bigcap_{i \in S} A(i)$. EJR requires some $i \in S$ with $|A(i) \cap R(E)| \geq |T|$.

Definition 4.3 (EJR, general additive)**.** S is (α, T)-*cohesive*, for $\alpha\colon C \to [0,1]$ and $T \subseteq C$, if $|S|/n \geq \mathrm{cost}(T)/b$ and $u_i(c) \geq \alpha(c)$ for all $i \in S$, $c \in T$. A rule satisfies *EJR* if for every such S there is $i \in S$ with $u_i(R(E)) \geq \sum_{c \in T} \alpha(c)$.

Key idea

A group that is an $|S|/n$ fraction of voters and can name an affordable bundle T they all value is "owed" the utility of T—and EJR guarantees *at least one* member actually gets it. The "at least one" is not a weakness: requiring it for *every* member is impossible even on tiny instances (Aziz et al., 2017).

Watch out — a subtle point that is easy to get wrong

EJR is genuinely demanding here. With a single voter, an EJR outcome must solve the *knapsack* problem; since knapsack is (weakly) NP-hard, *no strongly polynomial rule can satisfy EJR exactly in the general additive model* (Peters et al., Prop. 1). Equal Shares is polynomial, so it cannot satisfy EJR exactly there—only "up to one project."

Definition 4.4 (EJR up to one project)**.** R satisfies *EJR up to one project* if for every (α, T)-cohesive S there is $i^* \in S$ such that either $u_{i^*}(R(E)) \geq \sum_{c \in T} \alpha(c)$, or for some $c^* \in T$, $u_{i^*}(R(E) \cup \{c^*\}) > \sum_{c \in T} \alpha(c)$.

Theorem 4.2 (Peters et al., 2021)**.** *The Method of Equal Shares satisfies EJR up to one project in the general PB model. In the approval-based model (any costs) the "up to one" clause never bites, so* MES *satisfies EJR* exactly*.*

Proof idea (the three-runs argument). Fix an (α, T)-cohesive S; set $\alpha_c = \min_{i \in S} u_i(c)$ and $\alpha = \sum_{c \in T} \alpha_c$. Compare three runs of MES: (A) the real run, output W; (B) the run in which voters of S face *no* budget cap when buying projects of T; (C) a shrunken run with only S and only T, unlimited budgets, and $u_i(c) = \alpha_c$. In (C) each $c \in T$ is σ_c-affordable with $\sigma_c = \mathrm{cost}(c)/(|S|\alpha_c)$. Track, for a fixed voter i^*, the money $f_{(B)}(x)$ and $f_{(C)}(x)$ needed to reach utility x. One shows $f_{(C)}(x) \geq f_{(B)}(x)$ (the slope of $f_{(C)}$, namely σ_{c_s}, dominates that of $f_{(B)}$, because in (B) there are extra payers and weakly higher utilities). Since $f_{(C)}(\alpha) = \mathrm{cost}(T)/|S| \leq b/n$, we get $f_{(B)}(\alpha) \leq b/n$: by the time i^* has spent her whole share she has reached utility α (giving the exact bound), or overshoots on one extra project $c^* \in T$ (the "up to one" case). □

Corollary 4.1 (District proportionality, the easy case)**.** *If every voter in district d approves exactly the projects of d, they form a T-cohesive group for T any affordable bundle of d's projects with* $\mathrm{cost}(T) \leq \pi_d b$*. By Theorem 4.2,* MES *funds at least $|T|$ of them for some member—so district d controls at least its population share π_d of the budget.*

Connection to Durango

Corollary 4.1 is the heart of the Durango argument. The DEM *imposes* $b_d \approx \pi_d\, b$ from above (a quota). MES *produces* the same π_d floor *endogenously*, while *also* representing non-geographic groups (parents, cyclists) and never forcing an uninformed vote. We name this guarantee *π_d-geographic equity* and develop it in §6.

4.3 Stronger and weaker neighbours: the JR lattice

Definition 4.5 (The core)**.** W is in the *core* if for every $S \subseteq N$ and $T \subseteq C$ with $|S|/n \geq \mathrm{cost}(T)/b$ there is $i \in S$ with $u_i(W) \geq u_i(T)$.

The core lets a group propose *any* affordable counter-bundle (not just a unanimously liked one), so it implies EJR. It may be *empty*, even with unit costs (Peters et al., Example 2), which is why EJR—restricting to cohesive T—is the workable target. Equal Shares nonetheless *approximates* the core:

Theorem 4.3 (Peters et al., 2021)**.** *With $u_{\max}/u_{\min}$ the ratio of the largest to smallest positive attainable voter utilities, the Equal-Shares outcome lies in the α-core for $\alpha = 4\log(2\,u_{\max}/u_{\min})$.*

Definition 4.6 (Priceability; Peters and Skowron, 2020)**.** W is *priceable* if there is a budget $b' \geq b$ (each voter holding b'/n) and payments $p_i\colon C \to \mathbb{R}_{\geq 0}$ with: (C1) $u_i(c) = 0 \Rightarrow p_i(c) = 0$; (C2) $\sum_c p_i(c) \leq b'/n$; (C3) each $c \in W$ is fully paid; (C4) nothing is paid for $c \notin W$; (C5) for each $c \notin W$ the unspent money of its supporters is $\leq \mathrm{cost}(c)$. Equal Shares is always priceable—it builds the price system as it runs.

Definition 4.7 (Exhaustiveness; Aziz et al., 2018)**.** R is *exhaustive* if no unfunded project fits in the remaining budget: $\mathrm{cost}(R(E) \cup \{c\}) > b$ for every $c \notin R(E)$.

Watch out — a subtle point that is easy to get wrong

Equal Shares is *not* exhaustive: a project can remain affordable in the aggregate while its own supporters are individually too poor to buy it, and the rule then declines to fund it. Moreover exhaustiveness is *incompatible* with priceability in general (Peters et al., Example 3). Completion methods (next) restore spending when that is desired.

Completion methods (for the leftover budget).

- *Varying the budget:* rerun MES with a gradually increased virtual budget $b' \geq b$, stopping just before real spending would exceed b. Keeps equal voting power and priceability; mirrors the d'Hondt divisor method.
- *Perturbation:* set tiny $u_i(c) = \varepsilon$ wherever utility was 0; with all utilities positive, MES is exhaustive (Peters et al., Prop. 2). Variant: $\varepsilon \cdot \mathrm{cost}(c)$, which experiments suggest is preferable.
- *Utilitarian greedy:* spend the remainder on unfunded projects by total utility $\sum_i u_i(c)$—this mimics the plurality rule cities already use.

4.4 Fully Justified Representation (FJR)

Definition 4.8 (FJR; Peters et al., 2021)**.** S is *weakly* (β, T)*-cohesive* if $|S|/n \geq \text{cost}(T)/b$ and $u_i(T) \geq \beta$ for every $i \in S$. A rule satisfies *FJR* if for each such S some $i \in S$ has $u_i(R(E)) \geq \beta$.

FJR weakens cohesion (the bundle T need not be unanimously liked project-by-project, only valued *overall* by each member), so it sits strictly between EJR and the core:

$$\text{JR} \subset \text{PJR} \subset \text{EJR} \subset \text{FJR} \subset \text{core}.$$

Both PAV and Equal Shares *fail* FJR (Peters et al., Examples 4–5). FJR is nonetheless satisfiable, by the *Greedy Cohesive Rule*:

Definition 4.9 (Greedy Cohesive Rule (GCR))**.** Start with $W = \varnothing$ and all voters active. Repeatedly find a value $\beta > 0$, an active group S, and $T \subseteq C \setminus W$ with S weakly (β, T)-cohesive; among these pick the one maximizing β (ties to smaller $\text{cost}(T)$). Add T to W, deactivate S. Stop when no such triple exists.

Theorem 4.4 (Peters et al., 2021)**.** *GCR satisfies FJR. Its proof uses only monotonicity of utilities, so GCR satisfies FJR even for general monotone (non-additive) valuations.*

Watch out — a subtle point that is easy to get wrong

GCR is custom-built for FJR and pays for it: it is *not* polynomial, *not* priceable or exhaustive on its own (though every GCR outcome can be *completed* to a priceable one), and it can violate *laminar proportionality* that Equal Shares respects (Peters et al., Examples 6–7). For Durango, Equal Shares—EJR-satisfying, polynomial, priceable—remains the rule to propose; FJR/GCR is the theoretical ceiling we cite, not the rule we recommend.

5 The DEM's distortions, in this framework

We now recast our five Durango distortions against the EJR baseline. (Full proofs live in `notebook_v2`; here we state each crisply in Peters et al.'s notation and flag where the foundation sharpens it.) Throughout, X^{DEM} and X^{UAM} denote funded sets under the two mechanisms.

Proposition 5.1 (Truncation welfare loss)**.** *Because* DEM *forces all* v *votes into one chosen district, a voter cannot express value for projects outside it. There exist profiles where* X^{DEM} *is strictly Pareto-dominated in utilitarian welfare* $\sum_i u_i(\cdot)$ *by an unconstrained outcome—boundary and city-wide projects (Circleville's* A, P*) are the canonical losers.*

Proposition 5.2 (Noise injection via κ_i)**.** *Suppose voter* i *values only projects in* κ_i *but must cast* v *votes in district* d_i*. Decompose a project's vote total as* $V(c) = \mathit{signal}(c) + \eta(c)$*, where* $\eta(c)$ *collects votes from voters with* $c \notin \kappa_i$*. Then* $\mathbb{E}[\eta(c)]$ *is independent of* c*'s quality, and for small/low-salience projects* η *can dominate signal—so vote totals are a biased proxy for value precisely where the budget is tightest.*

Proposition 5.3 (Coordination game)**.** *District* selection *under* DEM *is a strategic stage: a voter's optimal district depends on where others go (to clear within-district funding thresholds). The induced game has multiple Nash equilibria selecting different X^{DEM}; focal points (salience s_d) drive selection.*

Proposition 5.4 (Mobilization advantage)**.** *An organized institution (e.g. a university) acts as an equilibrium selector, steering its members into one district and displacing roughly $v \cdot |\mathit{members}|$ votes from those members' otherwise-preferred districts—an advantage unrelated to project quality.*

Proposition 5.5 (Equity tradeoff: why the constraint exists)**.** *Under* UAM*, salience dominates: with $s_{d_H} > s_{d_L}$, low-salience district d_L can receive* zero *funding. The* DEM*'s per-district budgets b_d prevent this—so the constraint, for all its faults, is an* accidental equity instrument*. Any reform must keep the π_d floor it provides.*

Connection to Durango

Propositions 5.1–5.5 are exactly Peters et al.'s informal Circleville complaints turned into propositions, plus two Durango-specific ones (κ_i noise, mobilization) that their approval model does not foreground. The equity tradeoff (5.5) is the pivot: it forbids the lazy reform "just drop the constraint" (= UAM) and forces us to a rule that keeps the floor *endogenously*.

6 How Equal Shares repairs each distortion

Proposition 6.1 (No truncation)**.** *Under* MES *there is a single city-wide ballot: a voter assigns utility to* any *project, so boundary and city-wide projects compete on equal footing. The truncation of Prop. 5.1 cannot arise.*

Proposition 6.2 (No forced noise)**.** *A voter pays only for projects with $u_i(c) > 0$, i.e. within κ_i. No mechanism pressure converts ignorance into votes, removing the $\eta(c)$ term of Prop. 5.2.*

Proposition 6.3 (Reduced coordination)**.** *There is no district to select, so the* DEM *selection game (Prop. 5.3) disappears.* MES *is still not strategyproof (below), but the multiplicity of district-selection equilibria is gone.*

Proposition 6.4 (Bounded mobilization)**.** *An institution can still concentrate utility, but* MES *charges each supporter from her own share b/n: a bloc of size k commands at most kb/n of spending. Mobilization advantage is* capped by group size*, not amplified by displacement as in Prop. 5.4.*

Theorem 6.1 (π_d-geographic equity)**.** ***(Coined name; result is a corollary of EJR.)*** *Under* MES*, if the residents of district d can name an affordable bundle of d's projects they all value, then d commands at least its population share π_d of the budget. Hence no district is zero-funded as it can be under* UAM *(Prop. 5.5)—the equity the* DEM *provided by quota is provided by* MES *by construction (Corollary 4.1).*

A name we are introducing (our label, not standard terminology)

We call the guarantee of Theorem 6.1 *π_d-geographic equity.* The *name* is ours; the *content* is Peters et al.'s EJR specialized to the district structure. Stating it as "equity by construction" is the paper's central selling point for the reform.

Proposition 6.5 (Honest limitation: MES is not strategyproof)**.** *There are profiles where a voter benefits from misreporting (concrete counterexample in `notebook_v2`).* ***The operative reason is not Gibbard–Satterthwaite****—GS needs an unrestricted preference domain, which PB does not have—but the EJR-style tension itself: a rule delivering proportional shares cannot also be strategyproof (cf. Peters & Skowron 2020 on the limits of welfarism). We report this openly rather than overclaim.*

Watch out — a subtle point that is easy to get wrong

This is the one place where the foundation *corrects* an earlier draft instinct: do not cite Gibbard–Satterthwaite for MES's non-strategyproofness. The PB domain is restricted; the right citation is the proportionality/welfarism literature (Peters & Skowron 2020), not GS.

7 Comparison: DEM vs. UAM vs. MES

Property	**DEM** (Durango now)	**UAM** (drop constraint)	**MES** (proposed)
City-wide ballot	No (one district)	Yes	Yes
Boundary / city-wide goods	Orphaned	OK	OK
Forces uninformed votes (κ_i)	Yes (v per district)	No	No
Geographic equity	By quota b_d	**None** (salience wins)	**By construction** (π_d, EJR)
Non-geographic groups	Silenced	Plurality only	Represented (EJR)
Proportionality axiom met	—	— (fails JR)	EJR (exact, approval)
Priceable	—	No	Yes
Polynomial-time	Yes	Yes	Yes ($O(m^2 n \log n)$)
Strategyproof	No	No	No (Prop. 6.5)

Key idea

UAM fixes DEM's fragmentation but destroys equity; DEM protects equity but fragments and adds noise. MES is the only column that is "Yes/OK/By construction" down the equity-and-voice rows while staying polynomial—which is the whole argument for the reform.

8 Toward the paper: the simulation *(stub — to expand)*

This section is intentionally a stub in the first cut. It records the design so the build order is fixed; the full treatment comes once we lock the data model.

Planned, mirroring Peters et al.'s experiments (their §6) but on Durango ballots:

- **Counterfactual.** Given the same ballots, compute X^{DEM} (actual) and X^{MES}; compare funded sets, utilitarian welfare, and the distribution of voter utility (Peters et al. find global rules give *more equal* utility and usually higher total than separate district elections).
- **Utility reading.** Run both *approval* and *cost* utilities (§2) as a robustness axis.
- **Completion.** Compare varying-budget vs. perturbation vs. utilitarian-greedy completions (§4).
- **Equity metric.** Report each district's funded share against π_d to show Theorem 6.1 bites empirically.
- **Synthetic ballots.** If individual ballots are unavailable, generate ballots consistent with published aggregates (votes per project, turnout per district).
- **Out of scope here:** CONAPO need-weighting and the ENIGH welfare link (follow-up).

A Notation crosswalk: `notebook_v2` → this notebook

Object	**old (v2)**	**new**	**note**
Voters	N	N	unchanged
Projects	$\mathcal{P}$	C (item c)	Peters; c is one project
Budget	B	b	Peters
Cost	c (macro)	$\mathrm{cost}(\cdot)$	frees c for a project
Per-voter votes (DEM)	v	v	unchanged
MES threshold	ϱ	ρ	dodge retired
EJR cohesion threshold	—	$\alpha(c)$	Peters
FJR weak-cohesion thresh.	—	β	Peters
District salience	α_d	s_d	frees α
Information function	σ_i	κ_i	frees σ (σ_c in EJR proof)
District population share	β_d	π_d	frees β
Signal / noise	$S(p), \eta(p)$	$S(c), \eta(c)$	only the project symbol changes

B Keystone references to add to `references.bib`

The load-bearing citations from Peters et al. for our paper are collected in `notebooks/refs_peters.bib` (BibTeX, ready to merge into `paper/references.bib`; the keys match the `\citet`/`\citealp` calls used above). The full list is rendered in the References below.

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Status. Tight first cut. Sections 1–7 are complete; Section 8 is a deliberate stub. Full proofs of Propositions 5.1–6.5 remain in `notebook_v2` and will be migrated into this notation as we expand.